Puneet Velidi

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Education

University of Victoria — Victoria, BC
MSc in Statistics (thesis-based, fully-funded) — *Jan. 2025 – Dec. 2026*
Co-supervisors: Prof. Farouk Nathoo & Prof. Michelle Miranda
Committee members: Tanya Garcia, Maryclare Griffin, and Enrico Amico | External examiner: Jörn Diedrichsen

Thesis: *Methods for Censored Covariates, Spatial Transcriptomic Data and Neural Manifolds*. Oral examination

Cornell University — Ithaca, NY
BA in Statistics & Economics — *Sep. 2018 – Dec. 2022*

Publications (†indicates joint first authorship)

Puneet Velidi, Maryclare Griffin, Juxin Liu, Brian Richardson, Jing Qian, Roland Matsouaka, Sujit Ghosh, Tanya Garcia, Farouk Nathoo. *SPARTACCUS: Semiparametric Bayesian Proportional Hazards Model with a Censored Covariate in Huntington's Disease.* **Submitted for publication**, 2026.

Puneet Velidi, Zhengxiao Wei, Farouk Nathoo. *Covariance Nonstationarity is Evident in Spatial Transcriptomics and Provides a New Categorization of Spatially Varying Genes.* **Under review**, 2026. bioRxiv

Puneet Velidi, Michelle Miranda, Farouk Nathoo, Ashery Mbilinyi, Cédric Beaulac. *Shape Analysis as an Inductive Bias for Glioma Grading from Tumor Contours.* **Submitted for publication**, 2026. arXiv

Puneet Velidi, Enrico Amico, Farouk Nathoo, Michelle Miranda. *Neural manifold connectomics reveals multiregime functional connectivity.* **Under review**, 2026. bioRxiv

Fabio Luchese, **Puneet Velidi**, . . . , Paolo Cassano. *Transcranial Photobiomodulation on Language and Cognitive Performance in Down Syndrome: A Pilot Randomized Sham-Controlled Trial.* **Under review**, 2026. medRxiv

B. Ferreras, . . . , **P. Velidi**, et al., P. Cassano. *Transcranial Photobiomodulation in Bipolar Disorder: A Case Series on Executive Skills and fMRI BOLD Signal.* **Submitted for publication**, 2026. SSRN

Guillermo Gonzalez-Garibay, **Puneet Velidi**, Akila Weerasekera, Nikos Makris, Carlos Lohmann, Katherine Anne Collins, Naomi Gaggi, Dan Vlad Iosifescu, Paolo Cassano. *Volumetric Changes Induced by Transcranial Photobiomodulation Therapy in Major Depressive Disorder: An Exploratory Neuroimaging Study.* **Submitted for publication**, 2026. SSRN preprint

Puneet Velidi†, Zhengxiao Wei†, Shreena N. Kalaria, Yimeng Liu, Céline M. Laumont, Brad H. Nelson, Farouk S. Nathoo. *Generalized Jeffreys's Approximate Objective Bayes Factor: Model-Selection Consistency, Finite-Sample Accuracy, and Statistical Evidence in 71,126 Clinical Trial Findings.* **Revise and resubmit (R&R)**, 2025. arXiv

S. Parker Singleton†, **Puneet Velidi**†, Louisa Schilling, Andrea I. Luppi, Keith Jamison, Linden Parkes, Amy Kuceyeski. *Altered structural connectivity and functional brain dynamics in individuals with heavy alcohol use elucidated via network control theory.* **Biological Psychiatry: Cognitive Neuroscience and Neuroimaging**, 2024. Link

Scholarships, Awards, Recognitions

2026 — **University of Victoria** — FGS Travel Award — *600 CAD*
Travel support for Bayesian Methods for the Social Sciences III at University College Dublin, Ireland, December 9–11, 2026; accepted poster, *Further developments of generalized Jeffreys's approximate Bayes factor.*

2026 — **ICSA-Canada Chapter** — Student/Postdoc Travel Scholarship Award — *600 CAD*
This award supports travel expenses for the International Chinese Statistical Association – Canada Chapter 2026 Symposium in Montreal.

2025 — **Labs4** — Technology Readiness Level-Up (TRL^) Fellowship — *15,000 CAD*
Lab-to-market grant; ChimeraML, combinatorial nanobody design with an ensemble of protein ML models for immune checkpoint inhibitor therapy. Funded by Mitacs, New Ventures BC, and BCIT. Labs4 TRL^ Pitch Competition, BC Region, *2nd place.*

2025 **University of Victoria** — Betty & Gilbert Kennedy Graduate Scholarship in Math. & Stats. *5,130 CAD*
Awarded to academically outstanding students in the Department of Mathematics and Statistics upon the recommendation by the department (*Donor Award*).

2025 **Maud Menten Institute/PIMS** — Research Accelerator Award *5,000 CAD*
This award supports a collaboration with the MGH Photobiomodulation Lab on Bayesian analysis of the TransPhoM-DS clinical trial.

2025 **University of Victoria** — FGS Graduate Fellowship (Master's) *5,000 CAD*
This fellowship is awarded to students of high academic standing upon recommendation by their department. *The typical M&S FGS Master's Fellowship is 2,500 CAD.*

2025 **University of Victoria** — FGS Travel Award & CUPE Conference Award *1,050 CAD*
These awards supported travel expenses for CNS 2025.

2025 **3MC–PIMS–IDMS–ICMS** — Summer School Supported Participant —
This recognition supported travel expenses + room and board for the two-week 2025 3MC–PIMS–IDMS–ICMS Summer School.

2025 **Maud Menten Institute/PIMS** — MMI HQP Summit Supported Participant —
This recognition supported travel expenses + room and board for the 2025 MMI HQP Summit.

Competitions

2026 **Y Combinator (P26)** —
Selected for an in-person interview in San Francisco out of 80k applications (approx. top 2.5%).

2026 **Critical Ops: A DC National Security Hackathon** (spon. OpenAI, Meta, Squadra Ventures) —
1st place out of 166 participants, ENEL (drone swarm 3D Gaussian Splat). Won demo opportunities with **OpenAI** and **Meta**.

2026 **SCSP AI+ Expo Hackathon** *5,000 USD*
2nd place out of 200+ participants, 1st place DC region, Electrical Grid Track. Franklin, IoT grid sound anomaly detection & agent-to-agent load negotiation.

2026 **Stanford University RNA 3D Folding Competition** —
Placed in the top 3% of 1,867 teams; **Silver Medal**.

Presentations

Oral Presentations

Brain Fingerprinting of Principal-component derived fMRI BOLD functional connectivity. The 7th ICSA-Canada Chapter Symposium, August **2026**. Montreal, QC. Invited Talk, **Invited Session Chair**.

A Bayesian Semiparametric Model for Survival Analysis with a Right Censored Covariate. 9th International Conference on Econometrics and Statistics, August **2026**. Kyoto, JP. Co-author of invited talk.

A Bayesian Semiparametric Model for Survival Analysis with a Right Censored Covariate. 2026 Statistical Society of Canada Annual Meeting, May **2026**. Hamilton, ON. Contributed Talk.

"Applications of Protein Folding Models in Immunotherapeutic Research." UBC Math Bio Seminar, January **2026**. Invited Seminar.

"70,000 Leagues Under the Sea of p-Values." Maud Menten Institute / Mathematical and Statistical Biology Seminar, October **2025**. Invited Seminar.

"70,000 Leagues Under the Sea of p-Values." UBC Stats Seminar, October **2025**. Invited Seminar.

Statistical methods for censored covariates. 2025 WNAR/IMS Annual Meeting, June **2025**. Whistler, BC. Co-author of discussant remarks for invited session.

"Network Control Theory in Neuroscience: Characterizing the Brain's Energy Landscape." Cognition and Brain Sciences Seminar, University of Victoria, March **2025**. Invited Seminar.

"Network Control Theory in Neuroscience: Characterizing the Brain's Energy Landscape." Maud Menten Institute / Mathematical and Statistical Biology Seminar, University of Victoria, February **2025**.

Poster Presentations

Further developments of generalized Jeffreys's approximate Bayes factor. Bayesian Methods for the Social Sciences III, University College Dublin, Ireland, December 9–11, **2026**. Accepted poster (upcoming).

Acute Effects of Transcranial Photobiomodulation On Glymphatic CSF Dynamics in Two Patients with Amnestic Mild Cognitive Impairment or Early Alzheimer Disease. Clinical Research Day, Mass General Brigham, October 30, **2026**. Virtual poster presentation.

A Bayesian Semiparametric Model for Survival Analysis with a Right Censored Covariate. The 7th ICSA-Canada Chapter Symposium, August **2026**. Montreal, QC.

Bayesian Analysis of the TransPhoM-DS Clinical Trial. Organization for Human Brain Mapping 2026 Annual Meeting, June **2026**. Bordeaux, FR.

Ahab's Harpoon: Calibrated irreplicability detection with approximate objective Bayes factors. 2026 Statistical Society of Canada Annual Meeting, May **2026**. Hamilton, ON.

Reassessing the Statistical Evidence in Clinical Trials with Extended Approximate Objective Bayes Factors. Maud Menten Institute 2025 HQP Summit and Working Group Meetings, July **2025**. Edmonton, AB.

Photobiomodulation therapy-induced volumetric brain changes on patients with depression. Cognitive Neuroscience Society 2025 Annual Meeting, March **2025**. Boston, MA.

Altered structural connectivity and functional brain dynamics in individuals with heavy alcohol use elucidated via network control theory. Organization for Human Brain Mapping 2024 Annual Meeting, June **2024**. Seoul, SK.

Network control analysis of individuals with substance use disorder. Organization for Human Brain Mapping 2023 Annual Meeting, July **2023**. Montreal, QC.

Research Experience

Graduate Researcher — Victoria, BC

University of Victoria — Department of Mathematics and Statistics — *Jan. 2025 – Present*
Nathoo Lab, PI: Prof. Farouk Nathoo & Prof. Michelle Miranda

Visiting Researcher — Birmingham, UK

University of Birmingham — School of Mathematics — *Aug. 2025*
AMICO Lab, PI: Prof. Enrico Amico

Postbacc Researcher — Boston, MA

Harvard Medical School & MGH — Division of Neuropsychiatry and Neuromodulation — *May 2024 – Dec. 2024*
MGH Photobiomodulation Laboratory, PI: Prof. Paolo Cassano

- Statistical analysis of TransPhoM-DS, TRIADE, TPEB, TRAP-AD clinical trials.

Undergraduate Researcher — Ithaca, NY

Weill Cornell Medicine — Department of Radiology — *Jan. 2022 – Dec. 2022*
Computational Connectomics Laboratory, PI: Prof. Amy Kuceyeski

Professional Experience

Founder Vancouver, BC & Washington, DC

ChimeraML — TRL Level-Up (Labs4) Fall 2025 Cohort *Sept. 2025 – Present*

- Combinatorial nanobody design using an ensemble of protein ML models for immune checkpoint inhibitor therapy. White paper available upon request.

Software Engineer II Bentonville, AR

Walmart Global Tech — EBS *Feb. 2023 – May 2024*

- *Early Career Recognition*, awarded to <10% of 2023 Walmart Global Tech Development Program.

AI/ML & NextGen Analytics Intern New York, NY

Memorial Sloan Kettering Cancer Center — DigITs *June 2022 – Aug. 2022*

Teaching Experience

Teaching Assistant Victoria, BC

University of Victoria — Department of Mathematics and Statistics *Jan. 2025 – April 2025*

Math Support Center Tutor Ithaca, NY

Cornell University — Department of Mathematics *Sep. 2021 – Dec. 2022*

Workshops and Short Courses

BIRS Workshop: A Roadmap towards Developing Mechanobiochemical Models for Single and Collective Cell Migration through Complex Non-Isotropic Environments. *Banff International Research Station (BIRS), Banff, AB.* May 10–15, **2026**.
Five-day invited workshop on 2- and 3-D mechanobiochemical models of single and collective cell migration.

eScience JASP Hackathon. *Netherlands eScience Center, Amsterdam, NL.* February 16–17, **2026**.
Developed a JASP module alongside the core JASP programming team.

3MC–PIMS–IDMS–ICMS Summer School on Quantitative Molecular and Cellular Biology. *University of Manitoba, Winnipeg, MB.* June 16–27, **2025**.
Two-week school on quantitative methods for structural, molecular, and cellular biology, using cancer as an integrative example.

Technical Reports

Computational Design of Cross-Reactive Nanobodies Targeting the Human Chemokine Receptor Family. ChimeraML, **2026**. *Available upon request.*

Immunodynamic Modeling in Melanoma and Gastric Cancer. Supervised by Kang-Ling Liao and Michael Chan. 3MC–PIMS–IDMS–ICMS Summer School on Quantitative Molecular and Cellular Biology, **2025**.

Professional Service

BMC Medical Research Methodology, Reviewer

Network Neuroscience, Reviewer

Human Brain Mapping, Reviewer (with Supervisor)

JASA Applications and Case Studies, Reviewer (with Supervisor)